

How Literacy and Age at Marriage Shape Family Size in Portugal, 1979*

A Statistic Research Using Poisson Regression Models

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February 2, 2025

TBD. IDK. Smash Them

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*Code and data are available at: https://github.com/Jie-jiao05/Family_Size.

1 Introduction

1.1 Why this topic is important

1.2 3 citation relate with our research

1.3 Our objective, goal, how the citation related to, what we gonna do

1.4 Brief introduction of our research

2 Method

2.1 Use GLM, possion, (reason), what is x how to interoperate

3 Result

3.1 Data Description

This dataset originates from the 1979-80 Portuguese Fertility Survey (Instituto Nacional de Estatística (Portugal) 1979), conducted as part of the World Fertility Survey (WFS) program and archived by Princeton University's Office of Population Research (OPR).

The Dataset includes 5,148 observations and 8 variables, covering factors such as literacy, marriage age, pregnancies, children, and regional demographics. Table 1 it shows a statistical summary of our dataset.

To analyze the relationship between family size, literacy, and age at marriage, the variable Literacy was categorized as a dummy variable, assigning values 0 and 1 to illiterate and literate individuals, respectively. The analysis primarily focused on demographic factors, economic influences, and regional differences, aiming to assess their impact on family size. Additionally, an offset variable accounting for the duration of marriage was included to control for varying exposure times.

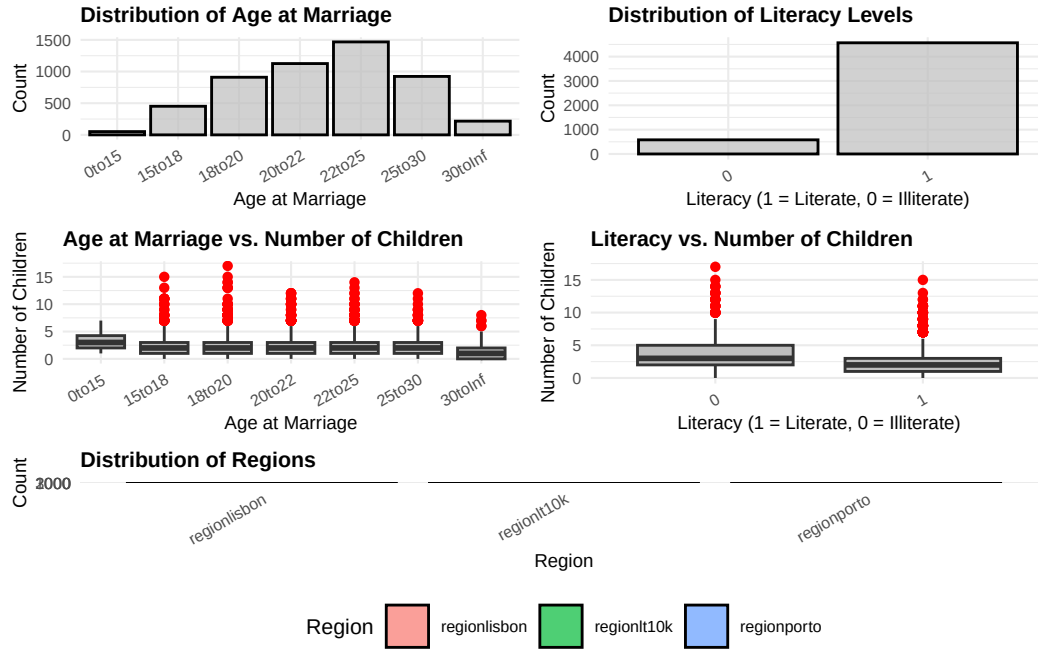


Figure 1: Distribution of Key Variable

Figure 1 illustrates the distribution of key variables and their relationship with family size. In general, the age of marriage is normally distributed within the range of this dataset, with a slight right skew, and the peak occurs at 22-25. At the same time, the Portuguese have a low level of education, most people come from regionlt10k, and literate people exist only as a very small minority in society. From the box plot of age at marriage and the number of children, we can see that the number of children is inversely proportional to age, and the lower the age of marriage, the higher the number of children. On the contrary, literate women have significantly more children than illiterate women.

3.2 Model Selection

Our Final model is showing above:

Table 1: Dataset Summary

X	age	ageMarried	monthsSinceM
Min. : 1	Min. :15.00	Length:5148	Min. : 0.0
1st Qu.:1288	1st Qu.:28.75	Class :character	1st Qu.: 73.0
Median :2574	Median :36.00	Mode :character	Median :148.0
Mean :2574	Mean :35.30		Mean :155.2
3rd Qu.:3861	3rd Qu.:43.00		3rd Qu.:230.0
Max. :5148	Max. :49.00		Max. :431.0
pregnancies	children	sons	literacy region10.20k
Min. : 0.000	Min. : 0.00	Min. : 0.000	0: 581 Min. :0.00000
1st Qu.: 1.000	1st Qu.: 1.00	1st Qu.: 0.000	1:4567 1st Qu.:0.00000
Median : 2.000	Median : 2.00	Median : 1.000	Median :0.00000
Mean : 2.282	Mean : 2.26	Mean : 1.187	Mean :0.08411
3rd Qu.: 3.000	3rd Qu.: 3.00	3rd Qu.: 2.000	3rd Qu.:0.00000
Max. :16.000	Max. :17.00	Max. :10.000	Max. :1.00000
region20k.	regionlisbon	regionlt10k	regionporto
Min. :0.0000	Min. :0.0000	Min. :0.0000	Min. :0.00000
1st Qu.:0.0000	1st Qu.:0.0000	1st Qu.:0.0000	1st Qu.:0.00000
Median :0.0000	Median :0.0000	Median :1.0000	Median :0.00000
Mean :0.1132	Mean :0.0913	Mean :0.6803	Mean :0.03108
3rd Qu.:0.0000	3rd Qu.:0.0000	3rd Qu.:1.0000	3rd Qu.:0.00000
Max. :1.0000	Max. :1.0000	Max. :1.0000	Max. :1.00000

$$\text{children}_i \sim \text{Poisson}(\lambda_i) \quad (1)$$

$$\log(\lambda_i) = -4.13839 - 0.12906 \cdot \text{literacy}_i \quad (2)$$

$$+ 0.03536 \cdot \text{ageMarried15to18}_i + 0.02006 \cdot \text{ageMarried18to20}_i \quad (3)$$

$$- 0.01574 \cdot \text{ageMarried20to22}_i - 0.03959 \cdot \text{ageMarried22to25}_i \quad (4)$$

$$- 0.02093 \cdot \text{ageMarried25to30}_i - 0.01054 \cdot \text{ageMarried30toInf}_i \quad (5)$$

$$- 0.06783 \cdot \text{region10_20k}_i - 0.20067 \cdot \text{region20k_}_i \quad (6)$$

$$- 0.18386 \cdot \text{regionlisbon}_i + 0.09381 \cdot \text{regionlt10k}_i \quad (7)$$

$$+ \text{NA} \cdot \text{regionporto}_i + \log(\text{monthsSinceM}_i) \quad (8)$$

3.3 Model selection (use model with offset)

3.4 Use likelyhood rario test to do the performance , to see the efficiency between model

3.5 Test overdispersion

4 Conclusion

Appendix

References

Instituto Nacional de Estatística (Portugal). 1979. *Portugal World Fertility Survey (WFS), 1979-80*. World Fertility Survey (WFS), Demographic; Health Surveys (DHS) Program.
<https://wfs.dhsprogram.com/index.cfm?ccode=pt>.